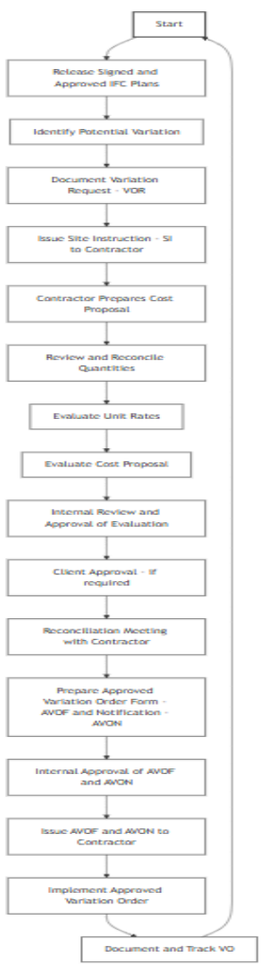
	Document No.	Section No.
	Effective Date:	Revision No.
	PROCEDURAL MANUAL	
	VARIATION ORDER	

OBJECTIVE	This procedure outlines the process for managing and controlling Variation Orders (VOs), also known as Change Orders, on BHBI construction projects. The aim is to ensure that all changes to the project scope, design, specifications, or schedule are properly documented, evaluated, approved, and implemented in a consistent and transparent manner. This procedure emphasizes the importance of Issued for Construction (IFC) plans as the baseline for variations, and incorporates the use of Site Instructions (SI), Cost Proposals, Approved Variation Order Forms (AVOF), and Approved Variation Order Notifications (AVON).
SCOPE	This procedure applies to all construction projects undertaken by BHBI and covers all types of project variations, including: • Client-requested changes • Design changes initiated by the Principal Designer or Project Engineer • Changes due to unforeseen site conditions • Changes required to comply with regulatory requirements • Value Engineering changes (as per the *Value Engineering Procedure*) • Subcontractor-requested changes

Responsibilities

- **Project Engineer:** Overall responsibility for managing the VO process on their projects. This includes reviewing, evaluating, recommending, and implementing VOs. The Project Engineer also assumes responsibilities typically associated with project management and construction management. Coordinates with the design team for issuing Site Instructions and obtaining approvals for variations.
- **Site Engineer:** Responsible for identifying potential variations on-site, documenting them, and assisting the Project Engineer in the VO process.
- **Principal Designer:** Responsible for reviewing and approving design-related VOs, and revising design documents as necessary. Provides technical guidance and support throughout the VO process.
- **Quantity Surveyor:** Responsible for assessing the cost impact of VOs, preparing cost estimates, reviewing and evaluating Cost Proposals from contractors, and tracking VO costs.
- **Procurement:** Responsible for managing the contractual and cost aspects of VOs related to subcontractors and suppliers.
- **Office Engineer:** Responsible for maintaining VO documentation (VORs, SIs, AVOFs, AVONs, etc.) and tracking the status of VOs in a Variation Order Register. Also provides administrative support to the VO process.
- **Client (or Client's Representative):** Responsible for reviewing and approving VOs, as required by the contract.
- **Contractor:** Responsible for submitting Cost Proposals for variations, as instructed by the Project Engineer through Site Instructions.

Flowchart



Procedure

1. Release of Issued for Construction (IFC) Plans:

- **Design Development:** The Principal Designer and design team develops the design to a level suitable for construction.
- **IFC Plan Preparation:** The Principal Designer and design team prepares IFC plans, incorporating all design details necessary for construction.
- **Signature and Approval:** IFC plans are signed by the Principal Designer and approved by the Project Engineer and any other designated BHBI authorities.
    - **Deliverables:** Signed and Approved IFC Plans

2. Variation Identification and Documentation:

- **Identify Potential Variation:** Any project team member, client representative, or subcontractor can identify a potential variation to the project. A variation can only proceed if there are signed and approved IFC Plans.
- **Document Variation Request:** The party identifying the variation must document it using a Variation Order Request (VOR) Form.
    - **Deliverables:** Variation Order Request (VOR) Form
- **Site Instruction (SI) Issuance:** If the Project Engineer determines that a variation is necessary, they issue a formal Site Instruction (SI) to the contractor. The SI provides detailed instructions regarding the changes in plans, designs, schedules, or scope.
    - **Deliverables:** Site Instruction (SI)
- **Contractor Acknowledgement:** The Contractor acknowledges receipt of the SI and confirms that they understand the instructions and will submit a Cost Proposal.

3. Cost Proposal Submission and Review:

- **Cost Proposal Preparation:** The Contractor prepares a detailed Cost Proposal based on the SI, including a breakdown of quantities, unit rates, and amounts for materials, labor, equipment, and any other costs.
    - **Deliverables:** Contractor's Cost Proposal
- **Cost Proposal Submission:** The Contractor submits the Cost Proposal to the Project Engineer within the timeframe specified in the SI or contract.
- **Quantity Reconciliation:** The Site Engineer and Quantity Surveyor review and reconcile the quantities in the Cost Proposal against the IFC plans and the SI.
- **Rate Evaluation:** The Quantity Surveyor evaluates the unit rates in the Cost Proposal, ensuring they are consistent with the contract rates (for additive or deductive variations) or justified by a Unit Price Analysis (UPA) for additional scopes.

4. Evaluation and Approval of Cost Proposal:

- **Cost Proposal Evaluation:** The Quantity Surveyor, in consultation with the Project Engineer, evaluates the Cost Proposal for accuracy, reasonableness, and compliance with the contract.
    - **Deliverables:** Cost Proposal Evaluation Report
- **Internal Review and Approval:** The evaluated Cost Proposal is reviewed by the Project Engineer and Principal Designer for approval of the evaluation and any comments regarding scoping and unit rates.
- **Client Approval (if required):** If required by the contract, the Project Engineer presents the evaluated Cost Proposal along with recommendations to the client for approval.

5. Variation Order Reconciliation and Issuance:

- **Quantity, Rate, and Amount Reconciliation:** The Project Engineer and Quantity Surveyor conduct a reconciliation meeting with the Contractor to discuss and agree upon the quantities, rates, and amounts for the variation.
    - **Deliverables:** Reconciliation Meeting Minutes
- **Approved Variation Order Form (AVOF) Preparation:** The Quantity Surveyor prepares the AVOF, detailing the breakdown of the original contract amount, previously approved variations, and the current variation, leading to the new total contract amount.
    - **Deliverables:** Approved Variation Order Form (AVOF)
- **Approved Variation Order Notification (AVON) Preparation:** The Project Engineer prepares the AVON, summarizing the changes to the contract amount and completion date, and formally authorizing the Contractor to proceed with the variation.
    - **Deliverables:** Approved Variation Order Notification (AVON)
- **Internal Approval:** The AVOF and AVON are reviewed by the Project Engineer, and approved by the Project Manager and any other designated BHBI authorities.
- **Issuance:** The Project Engineer issues the signed AVOF and AVON to the Contractor.

6. Variation Order Implementation:

- **Contract Modification:** Procurement modifies the project contract (with the client and/or subcontractors) to reflect the approved VO, if necessary.
    - **Deliverables:** Revised Contract Documents
- **Budget and Schedule Update:** The Quantity Surveyor updates the project budget and the Project Engineer updates the project schedule to reflect the approved VO.
    - **Deliverables:** Revised Project Budget, Revised Project Schedule
- **Design Revision (if applicable):** The Principal Designer revises the design documents to reflect the approved VO.
    - **Deliverables:** Revised Design Drawings, Revised Specifications
- **Construction Implementation:** The Project Engineer oversees the implementation of the approved VO on site, coordinating with the Site Engineer, subcontractors, and suppliers as needed.
- **Documentation:** The Office Engineer maintains a Variation Order Register to track all VORs, SIs, AVOFs, and AVONs, including their status, description, cost impact, and approval details. All VO documentation is properly filed and tracked.
    - **Deliverables:** Variation Order Register, Variation Order Files

Continuous Improvement:

- Regularly review and update this procedure to incorporate lessons learned and improve the efficiency of the variation order management process.

Records Generated (Deliverables)

- Signed and Approved IFC Plans

- Variation Order Request (VOR) Form
- Site Instruction (SI)
- Contractor's Cost Proposal
- Cost Proposal Evaluation Report
- Reconciliation Meeting Minutes
- Approved Variation Order Form (AVOF)
- Approved Variation Order Notification (AVON)
- Revised Contract Documents
- Revised Design Drawings
- Revised Specifications
- Variation Order Register
- Variation Order Files

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